# Exponential Growth and Decay Practice
## Answer Key
Algebra 1

### Quick Answers

1. 48
2. 64
3. $\frac{1}{2}$, 1, 2, 4, 8
4. $\frac{2}{9}$
5. 12800
6. 3
7. 12528.15
8. 12, $\frac{3}{4}$
9. 0
10. 12119
11. 5
12. 1954.67, 2130.63

### Curriculum

**Level:** Algebra 1
HSF-IF.A – *problems 1, 2, 4, 8*
HSF-IF.C – *problems 3, 6, 9, 11*
HSF-LE.A.2 – *problems 5, 7, 10, 12*
*Difficulty 1–5 of 5 across 18 tagged checks.*

### Common wrong answers

2. *If they got 160: multiplied by the base and then by the exponent instead of raising the base to the power*
7. *If they got 12.15: used the 15 percent lost as the multiplier instead of the 0.85 that is kept*
10. *If they got 11560: added 3 percent of the starting population twelve times instead of compounding*

**1.** $f(x) = 3 \cdot 2^x$**, find** $f(4)$**.** Substitute $x = 4$. The power comes first, the multiplication second: $2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$, and then $3 \cdot 16 = 48$. $\boxed{f(4) = 48}$

**2.** $g(x) = 100 \cdot (0.8)^x$**, find** $g(2)$**.** Here the base is less than 1, so this function decays. $(0.8)^2 = 0.8 \cdot 0.8 = 0.64$, and $100 \cdot 0.64 = 64$. $\boxed{g(2) = 64}$

**3. Table and graph of $h(x) = 2^x$.** Each value of $x$ is substituted into the same power. Reading left to right, a step of 1 in $x$ doubles the output — that constant ratio is what makes the curve exponential.

$h(-1) = 2^{-1} = \dfrac{1}{2^1}$, so $\boxed{h(-1) = \dfrac{1}{2}}$

$h(0) = 2^0$, and any nonzero base to the power 0 is one, so $\boxed{h(0) = 1}$

$h(1) = 2^1$, so $\boxed{h(1) = 2}$

$h(2) = 2 \cdot 2$, so $\boxed{h(2) = 4}$

$h(3) = 2 \cdot 2 \cdot 2$, so $\boxed{h(3) = 8}$

*Graph:* the five plotted points are $(-1, \frac{1}{2})$, $(0, 1)$, $(1, 2)$, $(2, 4)$, $(3, 8)$. The curve rises slowly on the left, passes through the $y$-axis at height one, and climbs steeply on the right; it never touches the $x$-axis. Accept any smooth curve through the five correct points.

**4. $p(x) = 2 \cdot 3^x$, find $p(-2)$.** A negative exponent means a reciprocal, not a negative answer: $3^{-2} = \dfrac{1}{3^2} = \dfrac{1}{9}$. Then $2 \cdot \dfrac{1}{9}$ gives the exact value. $\boxed{p(-2) = \dfrac{2}{9}}$

**5. $P(t) = 400 \cdot 2^t$ bacteria, find $P(5)$.** The starting count 400 is the value at $t = 0$; each hour multiplies by the base 2. After 5 hours the multiplier is $2^5 = 32$, so $400 \cdot 32 = 12800$ bacteria. $\boxed{P(5) = 12800}$

**6. $k(x) = 16 \cdot \left(\frac{1}{2}\right)^x$ litres — graph, then read off when 2 litres remain.** The table halves each time: $k(0) = 16$, $k(1) = 8$, $k(2) = 4$, $k(3) = 2$, $k(4) = 1$. Plotting those five points and joining them gives a curve falling toward the $x$-axis without ever reaching it. Reading across from height 2 meets the curve above $x = 3$, and the equation confirms it: $16 \cdot \left(\frac{1}{2}\right)^x = 2$ means $\left(\frac{1}{2}\right)^x = \dfrac{1}{8} = \left(\frac{1}{2}\right)^3$. $\boxed{x = 3}$

**7. $V(t) = 24000 \cdot (0.85)^t$ dollars, find $V(4)$.** Keeping 0.85 of the value each year means multiplying by 0.85 four times: $(0.85)^4 = 0.52200625$. Then $24000 \cdot 0.52200625 = 12528.15$ dollars. $\boxed{V(4) = 12528.15}$

**8. $f(x) = 6 \cdot (0.5)^x$, find $f(-1)$ and $f(3)$.** A negative exponent flips the base: $(0.5)^{-1} = \dfrac{1}{0.5} = 2$, so $6 \cdot 2$ gives $\boxed{f(-1) = 12}$

Going the other way, $(0.5)^3 = 0.125 = \dfrac{1}{8}$, so $6 \cdot \dfrac{1}{8} = \dfrac{6}{8}$, which reduces. $\boxed{f(3) = \dfrac{3}{4}}$

Notice the pattern: moving one step left multiplies by 2, moving one step right multiplies by 0.5.

**9. Graph $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$; find where they are equal.** The two curves are mirror images across the $y$-axis, because $\left(\frac{1}{2}\right)^x = 2^{-x}$. They can only meet where the exponent and its negative agree, and both functions pass through height one at the $y$-axis. Solving $2^x = 2^{-x}$ needs $x = -x$, which happens only at the origin's $x$-value. $\boxed{x = 0}$

**10. Town of** $8500$ **growing** $3$ **percent per year, population after** $12$ **years.** Growing by 3 percent multiplies by 1.03 once per year, so the model is $8500 \cdot (1.03)^t$ and $t = 12$ gives $(1.03)^{12} = 1.42576$ to five decimal places, and $8500 \cdot 1.42576 = 12118.97$, which rounds to a whole person. $\boxed{12119}$

**11.** $y = a \cdot 2^x$ **passes through** $(3, 40)$**; find** $a$**.** The point says that when $x = 3$ the output is 40, so substitute both: $a \cdot 2^3 = 40$. Since $2^3 = 8$, this is $8a = 40$. $\boxed{a = 5}$

The completed function is $y = 5 \cdot 2^x$, so the sketch passes through $(0, 5)$, $(1, 10)$, $(2, 20)$ and the given point $(3, 40)$.

**12. Account A:** $1200 \cdot (1.05)^t$**; Account B:** $900 \cdot (1.09)^t$**; compare after** $10$ **years.** Each rate becomes a multiplier: 5 percent growth is 1.05 per year and 9 percent growth is 1.09 per year.

$$1200 \cdot (1.05)^{10} = 1200 \cdot 1.628894627 = 1954.67$$
$$900 \cdot (1.09)^{10} = 900 \cdot 2.367363674 = 2130.63$$

$\boxed{A = 1954.67}$

$\boxed{B = 2130.63}$

Account B is worth more after 10 years even though it started with less: the larger base beats the larger starting value once enough years have passed.